# JCB MODEL I/O

**Input:**

**Output:**

## Quality Inspection Report: LEGO Technic Volvo Wheel Loader

**Status:** ✅ DEFECT-FREE / READY FOR DISPLAY

### Inspection Summary

All critical mechanical and structural components have been cross-referenced with the official Technic assembly diagrams. The vehicle has undergone a full functional and aesthetic check.

| Component | Status | Verification Detail |
|---|---|---|
| **Lifting Arm & Bucket** | ✅ Verified | The black lift arm is securely pinned, and the Volvo-branded bucket is level. |
| **Pneumatic/Mechanical Pins** | ✅ Verified | All blue and tan connector pins are fully inserted and flush with the beams. |
| **Wheel Assemblies** | ✅ Verified | All 4 wheels are securely attached to the axles with yellow hub rims visible. |
| **Steering Gear** | ✅ Verified | The top-mounted grey gear is centered and engages with the steering mechanism. |
| **Chassis Integrity** | ✅ Verified | The yellow and black frame is rigid with no loose beams or missing spacers. |

## Input:

## Output:

## Missing Part Alert: Front Loader Bucket

Your **LEGO Technic 30433 Volvo Wheel Loader** is almost complete, but the primary functional component is missing from the front assembly.

### 1. Identification

- **Part Name:** Large Technic Excavator Bucket.
- **Visual Marker:** The two black parallel lift arms at the front of the vehicle are currently exposed and empty.
- **Appearance:** A large, wide black piece shaped like a shovel or scoop

### 2. How to Confirm It is Missing

- Inspect the front of your build; while the black lift arms are present, they should be connected to a large bucket rather than ending with open connector holes.
- Compare your current build to the reference diagram where the bucket is shown as the final attachment for the front assembly.

### 3. Steps to Fix

- **Locate the Piece:** Find the large black bucket component in your remaining parts.
- **Attachment:** 1. Align the mounting holes on the back of the **Bucket** with the connector points at the end of the **parallel lift arms**. 2. Snap the bucket onto the pins until it is securely fastened to both arms.

- **Final Check:** Rotate the top gear of the loader to ensure the lift arms move the bucket up and down smoothly.

## Input:

## Output:

## Missing Part Alert: Front-Left Wheel Connector

Your **LEGO Technic 30433 Volvo Wheel Loader** is nearly finished, but a small grey connector is missing from the yellow steering arm.

### 1. Identification

- **Part Name:** Technic Bushing (Grey Connector).
- **Visual Marker:** The Right-angled L-shape (90°) yellow part next to the front-left tire is currently configured without its locking connector.
- **Appearance:** A small, round, ash-grey plastic cylinder with a hole in the middle.

### 2. How to Confirm It is Missing

- **Look at the Yellow Arm:** Find the Right-angled L-shape (90°) yellow piece located directly behind the front-left wheel.

- **Identify the Gap:** If you see the top hole of the yellow arm is empty and nothing is holding the vertical axle in place, the grey part is missing.

**3. Steps to Fix**

- **Step 1: Locate the Part** Find the small, ash-grey round bushing (the "screw" piece) in your remaining LEGO pieces.
- **Step 2: Clear the Assembly Area** Temporarily remove the yellow angled beam (the reversed L-shape) from the left side.
- **Step 3: Attach the Grey Connector** Insert the ash-grey connector directly into the **second hole** of the L-shape piece.
- **Step 4: Reinstall the Angled Beam** Place the reversed yellow angled beam back onto the assembly, ensuring it is now properly locked in place by the grey connector in that second position.